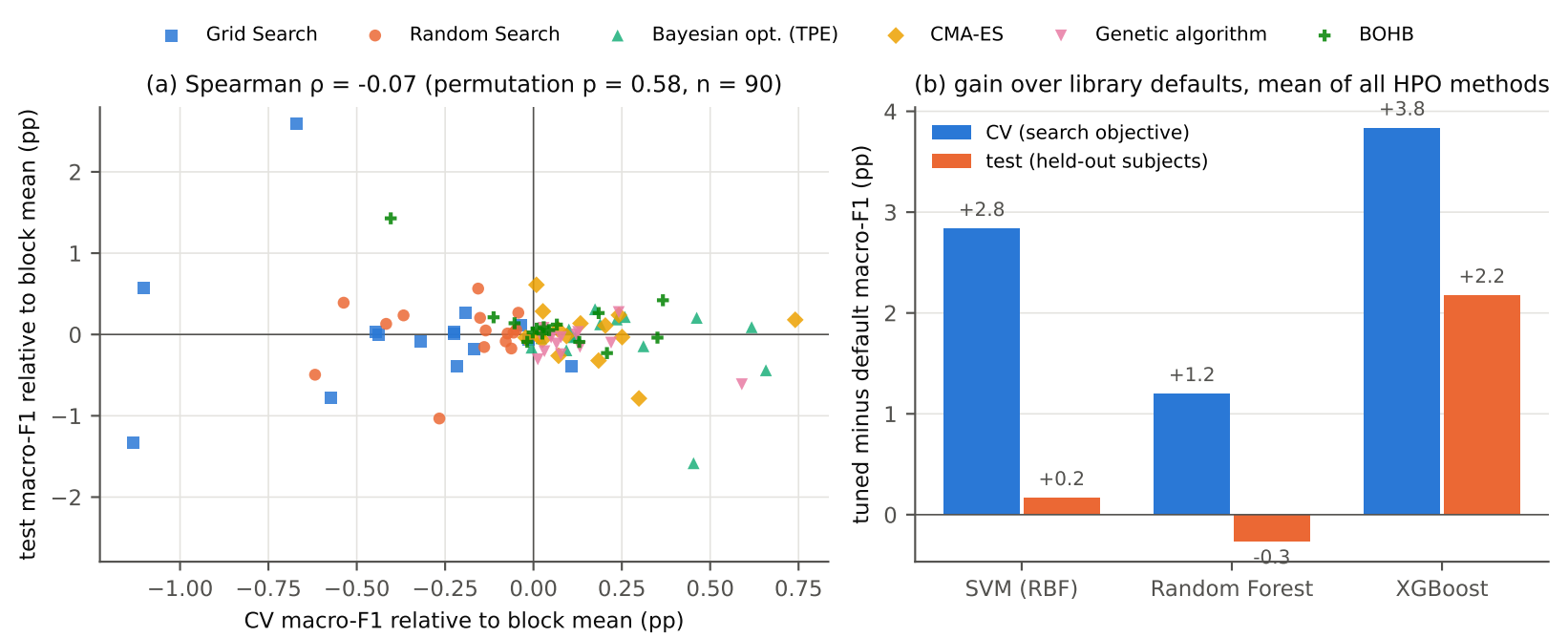
Grid Search
Random Search
Bayesian opt. (TPE)
CMA-ES
Genetic algorithm
BOHB
(a) Spearman ρ = -0.07 (permutation p = 0.58, n = 90)
test macro-F1 relative to block mean (pp)
CV macro-F1 relative to block mean (pp)
(b) gain over library defaults, mean of all HPO methods
tuned minus default macro-F1 (pp)
CV (search objective)
test (held-out subjects)
+2.8
+0.2
+1.2
-0.3
+3.8
+2.2
SVM (RBF)
Random Forest
XGBoost